Multimodal Prompt Template

Multimodal Prompt Template is a structured technique for combining user-typed questions, uploaded photos, and catalog images into one complete data package before sending it to the AI brain.
The function that communicates with OpenAI, the prompt sent can no longer be a simple paragraph of text, but rather an array called content_array. This list is arranged block by block like assembling Lego:

1. **Entering Text (Questions and Context)**
   The first block will always begin with text. The code will indirectly convey the core question to the user along with system instructions in an opening format:
   {"type": "text", "text": "..."}
   This code serves as a foundation of instructions about what the AI should analyze.

2. **Inserting User Uploaded Images (User Input Vision)**
   If I click the upload icon and send them a photo of their empty desk, then type, "What gadgets from the store would complement this desk?", the code will capture the image in the user_img_b64_resized variable. Then, a second block is added to the array:
   {"type": "image_url", "image_url": {"url": "data:image/jpeg;base64,..."}}
   Now, AI can see the user's work desk

3. **Insert Product Reference Image (Context Vision)**
   Context Vision is the most precise part and makes the assistant truly intelligent. The AI isn't just asked to look at user images, but is also coded to proactively send visual store brochures.
   - I created a code to loop through the top 3 relevant products using the for product in product_data command.
   - The previous code will look for the location of the product image file in the static folder.
   - The store's product images are added as additional image_url blocks to the prompt.

Vital Role Behind the Scenes: Base64 Minification & Conversion

OpenAI's API key does not accept RAW image file submissions, but only accepts JSON text format.

This function performs two heavy tasks automatically:

- **Resolution Reduction (Resizing):** If a user uploads a 5MB photo directly from their phone camera at 4K resolution, sending it directly to the AI would be very expensive, consuming many tokens and requiring a long response time. This function will shrink the image, for example, to a maximum of 400x400 pixels while maintaining its aspect ratio.
- **Base64 Conversion:** Once compressed, the image is converted into a very long text code called Base64 format. The encoded text string embedded in Base64 is the content_array so the image can be sent smoothly over internet protocols.

Comparison: Plain Text vs Multimodal

Component	Text Only (Single-modal)	Multimodal
User Input Form	Find me a maroon bag.	Upload a photo of a red bag and type: Find one exactly like this.
AI Understanding Levels	Have to guess the texture or type of maroon in question.	Able to see texture, zipper shape, and specific details from the photo.
Providing References	AI only refers to the text description from products.json.	AI compares user photos with physical product images from product_data.
Data Delivery Format	AI compares user photos with physical product images from product_data.	A collection of advanced Arrays containing Text and Image Ciphers (Base64).